6.

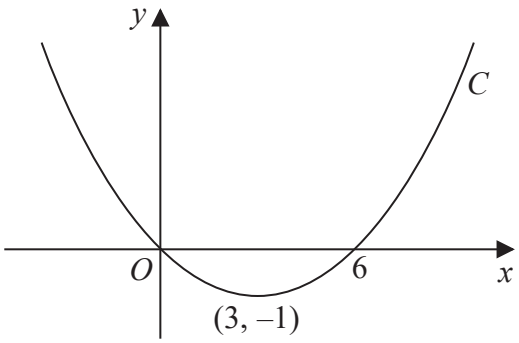


Figure 1

Figure 1 shows a sketch of the curve C with equation $y = f(x)$

The curve C passes through the origin and through $(6, 0)$

The curve C has a minimum at the point $(3, -1)$

On separate diagrams, sketch the curve with equation

(a) $y = f(2x)$

(3)

(b) $y = f(x + p)$, where p is a constant and $0 < p < 3$

(4)

On each diagram show the coordinates of any points where the curve intersects the x-axis and of any minimum or maximum points.